

Practical 2 - Understanding TinyOS Architecture and Execution Model

CODE:

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T110.py - C:/Users/IT-02/Desktop/T110 SANJANA SAHU/T110.py (3.14.6)
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print("T110 SANJANA SAHU")
import time
import queue
import threading

class TinyOSSchedulersSimulation:
    def __init__(self):
        # The core bounded FIFO task queue used by the TinyOS scheduler loop
        self.task_queue = queue.Queue(maxsize=8)
        self.is_running = True
        self.system_lock = threading.Lock()

    def post_task(self, task_name, computation_time):
        """Simulates the nesC 'post taskName()' command."""
        try:
            self.task_queue.put_nowait((task_name, computation_time))
            print(f"[SYSTEM]: Task '{task_name}' successfully posted to FIFO queue.")
        except queue.Full:
            print(f"[WARNING]: FIFO Task Queue Full! Task '{task_name}' dropped.")

    def trigger_hardware_event(self, event_name, bound_task_name, comp_time):
        """Simulates an asynchronous hardware interrupt signaling an event."""
        print(f"\n⚡ [HARDWARE INTERRUPT]: Event '{event_name}' fired!")
        # In TinyOS, an event handler responds instantly and posts a deferred task
        self.post_task(bound_task_name, comp_time)

    def run_scheduler_loop(self):
        """The main TinyOS execution loop."""
        print("[INIT]: TinyOS Scheduler Loop Initialized. Waiting for tasks...")

        while self.is_running or not self.task_queue.empty():
            try:
                # Emulate entering low-power sleep mode if queue is empty
                task_name, comp_time = self.task_queue.get(timeout=1)

                print(f"\n📡 [SCHEDULER]: Dispatching '{task_name}' (FIFO Front)...")

                # Non-preemptive execution block
                with self.system_lock:
                    start_time = time.time()
                    # Simulate processing computational workload
                    time.sleep(comp_time)
                    print(f"✅ [TASK COMPLETE]: '{task_name}' ran to completion in {comp_time}s.")

                self.task_queue.task_done()

            except queue.Empty:
                if not self.is_running:
                    break
                # TinyOS drops the CPU into ultra-low power state until a HW interrupt fires
                print("[POWER]: No pending tasks. MCU entering low-power sleep mode...")
                time.sleep(0.5)

# --- Test Execution Case ---
if __name__ == "__main__":
    simulation = TinyOSSchedulersSimulation()

    # Start the continuous TinyOS scheduler processing loop in a thread
    scheduler_thread = threading.Thread(target=simulation.run_scheduler_loop)
    scheduler_thread.start()
```

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    self.task_queue.task_done()

except queue.Empty:
    if not self.is_running:
        break
    # TinyOS drops the CPU into ultra-low power state until a HW interrupt fires
    print("[POWER]: No pending tasks. MCU entering low-power sleep mode...")
    time.sleep(0.5)

# --- Test Execution Case ---
if __name__ == "__main__":
    simulation = TinyOSSchedulersSimulation()

    # Start the continuous TinyOS scheduler processing loop in a thread
    scheduler_thread = threading.Thread(target=simulation.run_scheduler_loop)
    scheduler_thread.start()

    # Simulate random asynchronous hardware triggers (e.g., sensor timers or radios)
    time.sleep(0.5)
    simulation.trigger_hardware_event("Timer0.fired", "ReadSensorTask", 1.5)
    simulation.trigger_hardware_event("Radio.receive", "ProcessPacketTask", 0.8)

    # Intentionally fire an event mid-execution to observe non-preemption behavior
    time.sleep(0.4)
    simulation.trigger_hardware_event("ADC.dataReady", "FilterDataTask", 0.5)

    # Allow simulation to run through processing
    time.sleep(4.0)
    simulation.is_running = False
    scheduler_thread.join()
    print("\n[STOP]: Simulation Finished.")

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```

OUTPUT:

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IDLE Shell 3.14.6
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Python 3.14.6 (tags/v3.14.6:c63a6c6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/IT-02/Desktop/T110 SANJANA SAHU/T110.py =====
T110 SANJANA SAHU
[INIT]: TinyOS Scheduler Loop Initialized. Waiting for tasks...

⚡ [HARDWARE INTERRUPT]: Event 'Timer0.fired' fired!
[SYSTEM]: Task 'ReadSensorTask' successfully posted to FIFO queue.
📦 [SCHEDULER]: Dispatching 'ReadSensorTask' (FIFO Front)...

⚡ [HARDWARE INTERRUPT]: Event 'Radio.receive' fired!
[SYSTEM]: Task 'ProcessPacketTask' successfully posted to FIFO queue.

⚡ [HARDWARE INTERRUPT]: Event 'ADC.dataReady' fired!
[SYSTEM]: Task 'FilterDataTask' successfully posted to FIFO queue.
✅ [TASK COMPLETE]: 'ReadSensorTask' ran to completion in 1.5s.

📦 [SCHEDULER]: Dispatching 'ProcessPacketTask' (FIFO Front)...
✅ [TASK COMPLETE]: 'ProcessPacketTask' ran to completion in 0.8s.

📦 [SCHEDULER]: Dispatching 'FilterDataTask' (FIFO Front)...
✅ [TASK COMPLETE]: 'FilterDataTask' ran to completion in 0.5s.
[POWER]: No pending tasks. MCU entering low-power sleep mode...

[STOP]: Simulation Finished.

>>>

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```